Multiple Linear Regression: The Normal Equation

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Where this fits: Pipeline map step 8 (Model). See the Course map.

0 Foundations	1 Frame	2 Get data	3 Understand	4 Clean	5 Features	6 Reduce
7 Split	**8 Model**	9 Evaluate	10 Tune	11 Deploy	12 Test	13 Monitor

- **Builds on:** Linear combinations, span and basis; Ordinary least squares (closed form); Multiple linear regression; Jacobian and matrix gradients.
- **Leads to:** Ridge regression.
- **Compare with:** Moore-Penrose pseudo-inverse; Gradient descent.

1. Overview

Key point: With several features we still find the best coefficients the same way as with one: measure the total squared error, and find where its slope is zero in every direction. Matrices let us do that for every coefficient at once, and the answer is one formula, the **normal equation** (G-1344).

A **feature** (G-772) is an input variable (one column of the data table), the **target** (G-1949) is the output we predict, and an **observation** (G-1374) is one record (one row). Our running example predicts a student's package (the target) from CGPA (a feature).

For **simple linear regression** (G-1808), with one feature, we found two formulas, one for the **slope** (G-1823) and one for the intercept (see the two formulas). **Multiple linear regression** (G-1279) has several features (see the equation). Each feature gets its own weight, and there is one **intercept** (G-960) on top. These numbers are the **coefficients** (G-407). Writing a separate formula for each coefficient is hopeless when there are ten or a hundred features.

Matrices solve the problem. Like a spreadsheet formula dragged down a whole column instead of typed into every cell, a **matrix** (G-1180) lets us write one equation for all observations at once. We write all the data, all the predictions and all the coefficients as matrices. Then the same three steps as before give one formula for every coefficient at once:

1. write the error (Sections 2 and 3);
2. multiply it out and differentiate it (Sections 4 and 5);
3. set the derivative to zero and solve (Section 6).

The result, at the end of Section 6, is the **normal equation**. Section 6.1 runs it by hand on four students, with every sum and the **inverse** (G-968) worked out on paper. Our own class then codes it from scratch. Each rule about matrices is checked on small numbers before it is used.

A note on letters. In this Note the letter m counts the features (for example $m = 3$ for CGPA, IQ and gender). In the error function of simple linear regression, m was the slope. Here every slope is a coefficient, written β (the Greek letter beta).

2. The model in matrix form

> **Key point:** Put the data in one table with an extra first column of 1s, and the **coefficients** (G-407) in one column. One multiplication of the two then gives every prediction at once.

2.1 One equation per student

> **Key point:** Every student gets the same equation with the same **coefficients** (G-407); only the feature values change.

Suppose we predict a student's package from three **features** (G-772): CGPA, IQ and gender. The model multiplies each feature by its coefficient and adds the intercept:

$$\hat{y} = \beta_0 + \beta_1 \cdot \text{cgpa} + \beta_2 \cdot \text{iq} + \beta_3 \cdot \text{gender}$$

Once the four numbers β_0 to β_3 are known, we can predict the package of any new student. Training means finding them.

With 100 students there are 100 such equations. To tell the values apart we give each value two labels, the student and the feature:

$$x_{ij} = \text{the value of feature } j \text{ for student } i$$

$$x_{21} = \text{the CGPA (feature 1) of student 2}$$

$$\hat{y}_1 = \beta_0 + \beta_1 x_{11} + \beta_2 x_{12} + \beta_3 x_{13}$$
$$\hat{y}_2 = \beta_0 + \beta_1 x_{21} + \beta_2 x_{22} + \beta_3 x_{23}$$

and so on, down to $\hat{y}_{100}$. The coefficients are the same in every line.

In general, with n **observations** (G-1374) and m features (for example n = 100 students and m = 3 features), observation i has the prediction

$$\hat{y}_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_m x_{im}$$

and there are n such equations.

2.2 Stacking the equations

> **Key point:** The n equations split into a table of numbers, X, times a column of **coefficients** (G-407), β.

A **matrix** (G-1180) is a table of numbers, and a **vector** (G-2081) is a single column of numbers. They let us write the n equations as one.

Figure 1 does it for four students with one **feature** (G-772), CGPA (the students of Section 6.1). Watch three things move:

1. the four left sides stack into one vector, $\hat{y}$;
2. the numbers stack into one matrix, X: a column of 1s and a column of CGPAs;
3. the coefficients, which were repeated in every line, are written once, as the vector β.

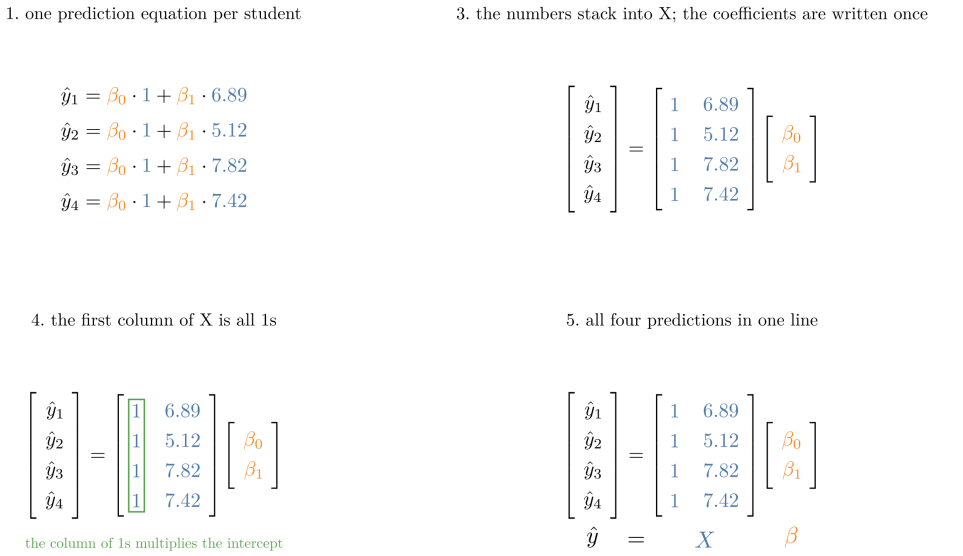


Figure 1: Four prediction equations become one matrix equation: the predictions stack into a vector, the numbers into the matrix X with a first column of 1s, and the coefficients are written once

Figure 2 shows the same split for n **observations** (G-1374) and m features.

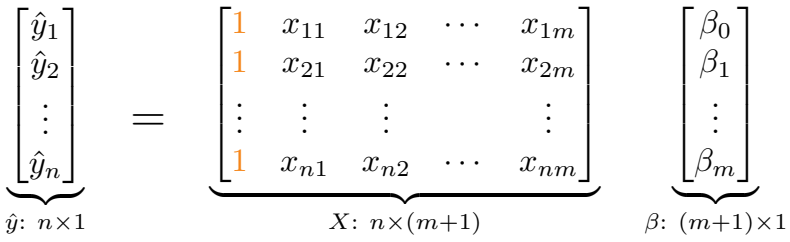


Figure 2: All predictions as one matrix product

- X, the **design matrix** (G-597), holds the data: one row per observation, one column per feature, plus a first column of 1s. The 1s multiply β_0, so the **intercept** (G-960) is treated like any other coefficient.
- β, the **coefficient vector** (G-410), holds the $m+1$ coefficients, β_0 to β_m.
- $\hat{y}$ holds the n predictions.

Multiplying one row of X by β means: multiply each entry of the row by the matching coefficient, then add. Row i gives

$$\beta_0 \cdot 1 + \beta_1 x_{i1} + \cdots + \beta_m x_{im}$$

which is exactly the prediction for observation i. So all predictions together are

$$\hat{y} = X\beta$$

With numbers: the first student of Figure 1 has the row [1, 6.89]. Section 6.1 finds the coefficients

$$\beta_0 = -0.811$$

$$\beta_1 = 0.565$$

The row times β, one product per line:

$$1 \times (-0.811) = -0.811$$

$$6.89 \times 0.565 = 3.893$$

$$-0.811 + 3.893 = 3.08$$

So 3.08 is the predicted package of that student.

> **Extra:** The shapes must fit: X is $n \times (m+1)$ and β is $(m+1) \times 1$. The inner sizes match, and the product is $n \times 1$, one prediction per row.

3. The error in matrix form

> **Key point:** Subtract each prediction from the real value to get one error per student. Square the errors and add them up: that one number measures how bad the **coefficients** (G-407) are.

For each student, the error is the real package minus the predicted one. This error on one **observation** (G-1374) is called a **residual** (G-705). The residuals of all observations form a **vector** (G-2081):

$$e = y - \hat{y} = y - X\beta$$

The **sum of squared errors** (G-1684) E adds up the squares of the residuals:

$$E = e_1^2 + e_2^2 + \cdots + e_n^2$$

That sum is a row times a column. The **transpose** (G-2012) of e, written e^{T}, is the same numbers written as a row (Figure 4 in Section 4 draws it). A row times a column multiplies matching entries and adds them, which is exactly the sum of squares:

$$E = e^{\mathsf{T}} e = (y - X\beta)^{\mathsf{T}} (y - X\beta)$$

With numbers, take three errors:

$$e = (0.2, -0.1, 0.3)$$

$$0.2 \times 0.2 = 0.04$$

$$(-0.1) \times (-0.1) = 0.01$$

$$0.3 \times 0.3 = 0.09$$

$$e^{\mathsf{T}}e = 0.04 + 0.01 + 0.09 = 0.14$$

Figure 3 draws the same idea on the four students of Section 6.1 and their best line. Each red segment is one entry of e, and the orange square on it has area e_i^2. Watch the biggest square: the student with CGPA 7.82 and error -0.36 supplies half of $E = 0.253$, because squaring makes large errors count much more than small ones.

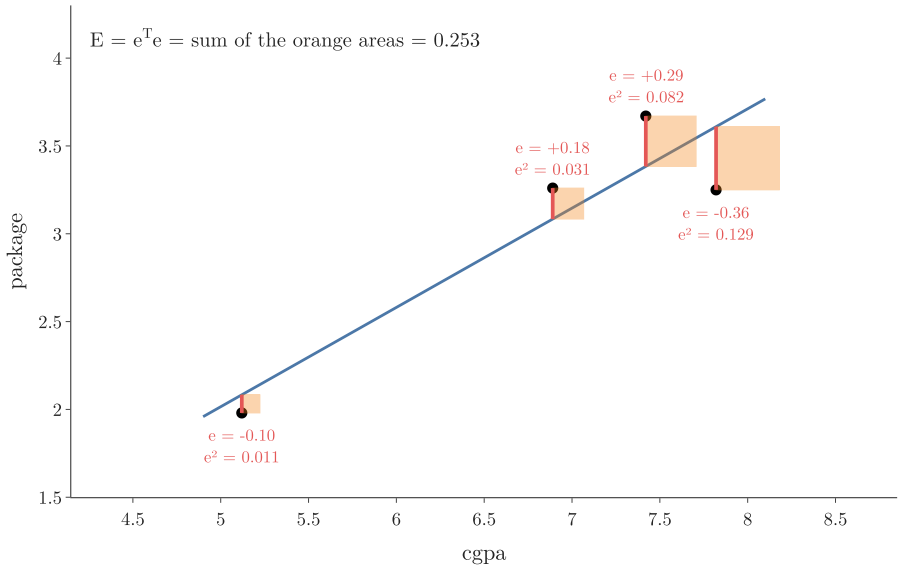


Figure 3: The four students of Section 6.1 and their best line: the residuals e (red) and their squares (orange); $E = e^{\mathsf{T}}e = 0.253$ is the total orange area

4. Expanding the error

> **Key point:** Multiplying out the brackets turns the error into three parts, exactly like $(a - b)^2 = a^2 - 2ab + b^2$ in ordinary algebra. Only two rules about flipping rows and columns are needed.

In ordinary algebra a square multiplies out into three parts:

$$(a - b)^2 = a^2 - 2ab + b^2$$

The error E is the **matrix** (G-1180) version of a square, and it multiplies out to the same three-part shape. To see every step on numbers, we use a tiny made-up example with two observations and one feature:

$$X = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix}$$

$$y = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$$

$$\beta = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

The first column of X is the column of 1s; the feature values are 2 and 3; the **targets** (G-1949) are 4 and 5; and we try the coefficients 1 and 1. The predictions and errors are

$$X\beta = \begin{bmatrix} 1 \times 1 + 2 \times 1 \\ 1 \times 1 + 3 \times 1 \end{bmatrix}$$

$$X\beta = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

$$e = y - X\beta = \begin{bmatrix} 4 - 3 \\ 5 - 4 \end{bmatrix}$$

$$e = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$E = e^{\mathsf{T}} e = 1 \times 1 + 1 \times 1 = 2$$

4.1 Two rules about transposes

> **Key point:** Transposing a product reverses its order, and a single number is its own **transpose** (G-2012).

Figure 4 shows both rules on the tiny example.

$$X\beta = \begin{bmatrix} 3 \\ 4 \end{bmatrix} \xrightarrow[\text{column} \to \text{row}]{\text{transpose}} (X\beta)^{\mathsf{T}} = \begin{bmatrix} 3 & 4 \end{bmatrix}$$

$$\beta^{\mathsf{T}} X^{\mathsf{T}} = \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 1+2 & 1+3 \end{bmatrix} = \begin{bmatrix} 3 & 4 \end{bmatrix}$$

same row: the transpose of a product is the product of the transposes, in reverse order

Figure 4: Top: the transpose turns the column Xβ = (3, 4) into the row [3, 4]. Bottom: multiplying the transposes in reverse order, β^{T}X^{T}, gives the same row

Rule 1: the transpose of a product reverses the order.

$$(AB)^{\mathsf{T}} = B^{\mathsf{T}} A^{\mathsf{T}}$$

Here A and B stand for any two matrices that can be multiplied. For our product, A is X and B is β, so the rule says

$$(X\beta)^{\mathsf{T}} = \beta^{\mathsf{T}} X^{\mathsf{T}}$$

Check on the tiny example. The left side flips the column (3, 4) into a row:

$$(X\beta)^{\mathsf{T}} = \begin{bmatrix} 3 & 4 \end{bmatrix}$$

The right side, with the transpose of X (its rows become columns):

$$X^{\mathsf{T}} = \begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}$$

$$\beta^{\mathsf{T}} X^{\mathsf{T}} = \begin{bmatrix} 1 \times 1 + 1 \times 2 & 1 \times 1 + 1 \times 3 \end{bmatrix}$$

$$\beta^{\mathsf{T}} X^{\mathsf{T}} = \begin{bmatrix} 3 & 4 \end{bmatrix}$$

The two sides match.

Rule 2: a single number is its own transpose. A 1×1 **matrix** (G-1180) has nothing to flip. The two terms below are each a single number, and each is the transpose of the other (by Rule 1), so they are equal:

$$y^\mathsf{T} X\beta = \beta^\mathsf{T} X^\mathsf{T} y$$

Check on the tiny example. The left side is the row of **targets** (G-1949) times the column of predictions:

$$y^\mathsf{T} X\beta = 4 \times 3 + 5 \times 4 = 12 + 20 = 32$$

For the right side, first multiply the transpose of X by y:

$$X^\mathsf{T} y = \begin{bmatrix} 1 \times 4 + 1 \times 5 \\ 2 \times 4 + 3 \times 5 \end{bmatrix}$$

$$X^\mathsf{T} y = \begin{bmatrix} 9 \\ 23 \end{bmatrix}$$

$$\beta^\mathsf{T} X^\mathsf{T} y = 1 \times 9 + 1 \times 23 = 32$$

Both are 32.

4.2 Multiplying out

> **Key point:** Four products come out, and the two middle ones are equal, so the error has three parts.

Multiply the two brackets of E term by term, as with (a − b)(a − b). Rule 1 turns the **transpose** (G-2012) of y − Xβ into the transpose of y minus the row from Figure 4:

$$(y - X\beta)^\mathsf{T} = y^\mathsf{T} - \beta^\mathsf{T} X^\mathsf{T}$$

The four products, one per line:

$$y^\mathsf{T} \cdot y = y^\mathsf{T} y$$

$$y^\mathsf{T} \cdot (-X\beta) = -y^\mathsf{T} X\beta$$

$$(-\beta^\mathsf{T} X^\mathsf{T}) \cdot y = -\beta^\mathsf{T} X^\mathsf{T} y$$

$$(-\beta^\mathsf{T} X^\mathsf{T}) \cdot (-X\beta) = +\beta^\mathsf{T} X^\mathsf{T} X\beta$$

Adding them:

$$E = y^\mathsf{T} y - y^\mathsf{T} X\beta$$

$$- \beta^\mathsf{T} X^\mathsf{T} y + \beta^\mathsf{T} X^\mathsf{T} X\beta$$

By Rule 2 the two middle terms are equal, so they combine:

$$E = y^\mathsf{T} y - 2y^\mathsf{T} X\beta + \beta^\mathsf{T} X^\mathsf{T} X\beta$$

Check on the tiny example, one part per line. The first part:

$$y^\mathsf{T} y = 4 \times 4 + 5 \times 5 = 41$$

The middle part, using the 32 of Rule 2:

$$2y^\mathsf{T} X\beta = 2 \times 32 = 64$$

The last part is the column Xβ = (3, 4) times its own transpose:

$$\beta^\mathsf{T} X^\mathsf{T} X\beta = (X\beta)^\mathsf{T} (X\beta)$$

$$= 3 \times 3 + 4 \times 4 = 25$$

Together:

$$E = 41 - 64 + 25 = 2$$

the same 2 as the direct **sum of squared errors** (G-1684) above.

5. Setting the derivative to zero

> **Key point:** At the best coefficients the error bowl is flat in every direction: its slope along each coefficient is zero. Two small rules turn that into one **matrix** (G-1180) equation.

As in **simple linear regression** (G-1808), the best coefficients sit at the bottom of the error bowl, where the slope is zero. Now there is one slope per coefficient. Figure 5 shows the two slopes on the four students of Section 6.1. Each panel cuts the error bowl along one coefficient while the other is held fixed, and the orange tangent shows the **slope** (G-1823)

there. Watch both slopes shrink as β walks to $(-0.81, 0.57)$: there both tangents lie flat at the same moment.

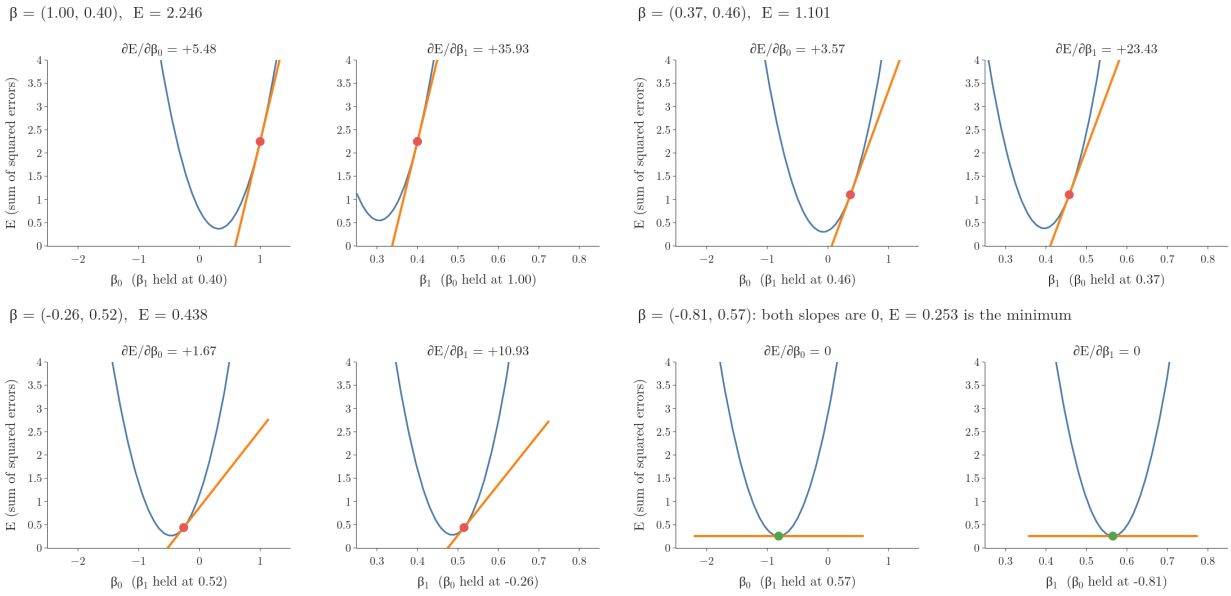


Figure 5: The gradient $-2X^\mathsf{T}y + 2X^\mathsf{T}X\beta$ on the four students: each panel is the error E along one coefficient, the orange tangent's slope is one entry of the gradient, and both reach 0 together at $\beta = (-0.81, 0.57)$

The slope along one **coefficient** (G-407), with the others held fixed, is a **partial derivative** (G-1457). Stacking all of them into one column gives the **gradient** (G-863). With two coefficients:

$$\frac{\partial E}{\partial \beta} = \begin{bmatrix} \partial E/\partial \beta_0 \\ \partial E/\partial \beta_1 \end{bmatrix}$$

"Both tangents flat" in Figure 5 means every entry of this column is 0.

5.1 Three rules of matrix calculus, each checked on numbers

> **Key point:** The **matrix** (G-1180) rules are the old rules "the derivative of a constant is 0", "of aβ is a" and "of aβ^2 is 2aβ", applied to every coefficient at once.

Rules for differentiating with **vectors** (G-2081) and matrices are called **matrix calculus** (G-1176). E has three parts, and each needs one rule. We check each rule by writing the part out for two **coefficients** (G-407), β_0 and β_1, taking the two ordinary **partial derivatives** (G-1457), and comparing.

Rule A: a part without β has derivative 0. The first part, $y^\mathsf{T}y$, is 41 in the tiny example whatever β is. A constant does not change when β changes, so

$$\frac{\partial}{\partial \beta}\left(y^{\mathsf{T}} y\right) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Rule B: a weighted sum of the coefficients has the weights as its derivative. The matrix version of "the derivative of aβ is a". Section 4.1 showed that the middle part is $\beta^{\mathsf{T}} X^{\mathsf{T}} y$, and that $X^{\mathsf{T}} y = (9, 23)$ in the tiny example. So, written out,

$$\beta^{\mathsf{T}} X^{\mathsf{T}} y = 9\beta_0 + 23\beta_1$$

The two partial derivatives, one per line:

$$\frac{\partial}{\partial \beta_0}(9\beta_0 + 23\beta_1) = 9$$

$$\frac{\partial}{\partial \beta_1}(9\beta_0 + 23\beta_1) = 23$$

Stacked, they are (9, 23), which is $X^{\mathsf{T}} y$ itself. In general:

$$\frac{\partial}{\partial \beta}\left(\beta^{\mathsf{T}} X^{\mathsf{T}} y\right) = X^{\mathsf{T}} y$$

and so, since the middle part is twice this (Rule 2):

$$\frac{\partial}{\partial \beta}\left(2 y^{\mathsf{T}} X \beta\right) = 2 X^{\mathsf{T}} y$$

Rule C: a "squared" part gives two times the matrix times β. The matrix version of "the derivative of aβ^2 is 2aβ". The last part has the square matrix $X^{\mathsf{T}} X$ in the middle. In the tiny example:

$$X^{\mathsf{T}} X = \begin{bmatrix} 1 \times 1 + 1 \times 1 & 1 \times 2 + 1 \times 3 \\ 2 \times 1 + 3 \times 1 & 2 \times 2 + 3 \times 3 \end{bmatrix}$$

$$X^{\mathsf{T}} X = \begin{bmatrix} 2 & 5 \\ 5 & 13 \end{bmatrix}$$

Written out, $\beta^{\mathsf{T}} X^{\mathsf{T}} X \beta$ has one term per entry of the matrix:

$$\beta^{\mathsf{T}} X^{\mathsf{T}} X \beta = 2\beta_0\beta_0 + 5\beta_0\beta_1$$

$$+ 5\beta_1\beta_0 + 13\beta_1\beta_1$$

Collecting the two middle terms, and calling this part P:

$$P = 2\beta_0^2 + 10\beta_0\beta_1 + 13\beta_1^2$$

Its two partial derivatives:

$$\frac{\partial P}{\partial \beta_0} = 4\beta_0 + 10\beta_1$$

$$\frac{\partial P}{\partial \beta_1} = 10\beta_0 + 26\beta_1$$

Now the matrix rule's answer, two times the matrix times β:

$$2X^\mathsf{T}X\beta = 2\begin{bmatrix} 2\beta_0 + 5\beta_1 \\ 5\beta_0 + 13\beta_1 \end{bmatrix}$$

$$2X^\mathsf{T}X\beta = \begin{bmatrix} 4\beta_0 + 10\beta_1 \\ 10\beta_0 + 26\beta_1 \end{bmatrix}$$

The same two lines. In general:

$$\frac{\partial}{\partial \beta}\left(\beta^\mathsf{T}X^\mathsf{T}X\beta\right) = 2X^\mathsf{T}X\beta$$

The rule works because $X^\mathsf{T}X$ is a **symmetric matrix** (G-1932): it equals its own **transpose** (G-2012), so the two 5s above are equal and add up to the 10 (MML §5.5).

> **Extra:** For any square matrix B the general rule is (MML eq. 5.107, which writes the gradient as a row):
>
> $$\frac{\partial(x^\mathsf{T}Bx)}{\partial x} = x^\mathsf{T}(B + B^\mathsf{T})$$
>
> With $B = X^\mathsf{T}X$, which equals its own transpose:
>
> $$B + B^\mathsf{T} = 2X^\mathsf{T}X$$

5.2 The gradient, then zero

> **Key point:** Adding the three derivatives gives the gradient; setting it to zero gives the **normal equations** (G-1344).

Apply Rules A, B and C to the three parts of E, one per line:

$$\frac{\partial}{\partial \beta}\left(y^{\mathsf{T}} y\right) = 0$$

$$\frac{\partial}{\partial \beta}\left(-2y^{\mathsf{T}} X\beta\right) = -2X^{\mathsf{T}} y$$

$$\frac{\partial}{\partial \beta}\left(\beta^{\mathsf{T}} X^{\mathsf{T}} X\beta\right) = 2X^{\mathsf{T}} X\beta$$

Adding them:

$$\frac{\partial E}{\partial \beta} = -2X^{\mathsf{T}} y + 2X^{\mathsf{T}} X\beta$$

Check on the tiny example at $\beta = (1, 1)$, where both errors were 1:

$$-2X^{\mathsf{T}} y = \begin{bmatrix} -18 \\ -46 \end{bmatrix}$$

$$2X^{\mathsf{T}} X\beta = \begin{bmatrix} 4 + 10 \\ 10 + 26 \end{bmatrix}$$

$$2X^{\mathsf{T}} X\beta = \begin{bmatrix} 14 \\ 36 \end{bmatrix}$$

$$\frac{\partial E}{\partial \beta} = \begin{bmatrix} -18 + 14 \\ -46 + 36 \end{bmatrix}$$

$$\frac{\partial E}{\partial \beta} = \begin{bmatrix} -4 \\ -10 \end{bmatrix}$$

The same **slopes** (G-1823) come from the simple-regression derivatives of the error function, minus two times the sum of the errors, and minus two times the sum of error times **feature** (G-772):

$$-2(1 + 1) = -4$$

$$-2(1 \times 2 + 1 \times 3) = -10$$

At the best **coefficients** (G-407) the gradient is zero:

$$-2X^\mathsf{T}y + 2X^\mathsf{T}X\beta = 0$$

Dividing by 2 and moving one term across:

$$X^\mathsf{T}X\beta = X^\mathsf{T}y$$

These are the **normal equations** (G-1345): m + 1 equations, one for each coefficient.

6. The normal equation

> **Key point:** To get the coefficients alone, we "divide" both sides by the **matrix** (G-1180) $X^\mathsf{T}X$. For matrices, dividing means multiplying by the inverse.

With ordinary numbers, "2 times a number is 6" is solved by dividing by 2, which is the same as multiplying by one half, the number that undoes 2. Matrices have no division; instead we multiply by the **inverse** (G-968), the matrix that undoes $X^\mathsf{T}X$. A matrix times its inverse gives the **identity matrix** (G-915), the matrix with 1s on the diagonal and 0s elsewhere, which changes nothing it multiplies.

For a 2×2 matrix the inverse has a three-step recipe (MML §2.2.2), shown in Figure 6:

1. swap the two numbers on the diagonal;
2. flip the sign of the other two;
3. divide everything by the diagonal product minus the off-diagonal product.

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \xrightarrow{\text{inverse}} \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

1. swap the two diagonal numbers
2. flip the sign of the other two
3. divide everything by $ad - bc$

$$\begin{bmatrix} 2 & 5 \\ 5 & 13 \end{bmatrix}^{-1} = \frac{1}{2 \times 13 - 5 \times 5} \begin{bmatrix} 13 & -5 \\ -5 & 2 \end{bmatrix} = \begin{bmatrix} 13 & -5 \\ -5 & 2 \end{bmatrix}$$

check: $\begin{bmatrix} 2 & 5 \\ 5 & 13 \end{bmatrix} \begin{bmatrix} 13 & -5 \\ -5 & 2 \end{bmatrix} = \begin{bmatrix} 26 - 25 & -10 + 10 \\ 65 - 65 & -25 + 26 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, the identity

Figure 6: The 2×2 inverse recipe, worked on the tiny example's $X^T X$ and checked: the matrix times its inverse is the identity

The number in step 3 is the **determinant** (G-598). For bigger matrices the computer does the work; the idea is the same.

On the tiny example (bottom of Figure 6), the determinant of the **transpose** (G-2012) of X times X is

$$2 \times 13 - 5 \times 5 = 26 - 25 = 1$$

so the inverse is the swapped matrix itself:

$$\begin{bmatrix} 2 & 5 \\ 5 & 13 \end{bmatrix}^{-1}$$

$$= \begin{bmatrix} 13 & -5 \\ -5 & 2 \end{bmatrix}$$

Multiply it by the column (9, 23) from Rule B:

$$\beta_0 = 13 \times 9 + (-5) \times 23 = 117 - 115 = 2$$

$$\beta_1 = (-5) \times 9 + 2 \times 23 = -45 + 46 = 1$$

The line "prediction $= 2 + 1 \times$ **feature** (G-772)" passes through both points:

$$2 + 1 \times 2 = 4$$

$$2 + 1 \times 3 = 5$$

In general, multiplying both sides of the **normal equations** (G-1345) by the inverse leaves β alone:

$$\beta = (X^{\mathsf{T}} X)^{-1} X^{\mathsf{T}} y$$

The formula is the **normal equation** (G-1344). A formula that gives the answer directly is a **closed-form solution** (G-398); this one gives all coefficients of **multiple linear regression** (G-1279) at once, and the method is called **ordinary least squares**, OLS (G-1406). scikit-learn's `LinearRegression` reaches the same solution with a numerically safer method than a literal inverse (scikit-learn docs, LinearRegression; LAPACK, DGELSD).

6.1 A worked example

> **Key point:** On four students with one feature, the **normal equation** (G-1344) gives the same slope and intercept as the simple formulas.

Take the first four students of the placement data, with CGPA 6.89, 5.12, 7.82, 7.42 and packages 3.26, 1.98, 3.25, 3.67.

1. **Build X:** a column of four 1s next to the four CGPAs.

$$X = \begin{bmatrix} 1 & 6.89 \\ 1 & 5.12 \\ 1 & 7.82 \\ 1 & 7.42 \end{bmatrix}$$

$$y = \begin{bmatrix} 3.26 \\ 1.98 \\ 3.25 \\ 3.67 \end{bmatrix}$$

2. **Compute the transpose of X times X.** Each entry is a column of X times a column of X. The column of 1s times itself counts the students:

$$1 + 1 + 1 + 1 = 4$$

   The column of 1s times the CGPAs adds the CGPAs:

$$6.89 + 5.12 + 7.82 + 7.42 = 27.25$$

The CGPAs times themselves add their squares:

$$47.4721 + 26.2144 + 61.1524 + 55.0564$$

$$= 189.8953$$

$$X^\top X = \begin{bmatrix} 4 & 27.25 \\ 27.25 & 189.8953 \end{bmatrix}$$

3. **Compute the transpose of X times y.** The column of 1s times y adds the packages:

$$3.26 + 1.98 + 3.25 + 3.67 = 12.16$$

The CGPAs times the packages, one product per student:

$$6.89 \times 3.26 = 22.4614$$

$$5.12 \times 1.98 = 10.1376$$

$$7.82 \times 3.25 = 25.4150$$

$$7.42 \times 3.67 = 27.2314$$

$$22.4614 + 10.1376 + 25.4150 + 27.2314$$

$$= 85.2454$$

$$X^\top y = \begin{bmatrix} 12.16 \\ 85.2454 \end{bmatrix}$$

4. **Invert with the recipe of Figure 6.** The **determinant** (G-598):

$$4 \times 189.8953 - 27.25 \times 27.25$$

$$= 759.5812 - 742.5625 = 17.0187$$

Swap the diagonal, flip the signs of the other two, divide by 17.0187:

$$(X^{\mathsf{T}}X)^{-1} = \frac{1}{17.0187}M$$

$$M = \begin{bmatrix} 189.8953 & -27.25 \\ -27.25 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 11.158 & -1.601 \\ -1.601 & 0.235 \end{bmatrix}$$

Check the top-left entry of the inverse times the **matrix** (G-1180), which should be 1:

$$\frac{189.8953 \times 4 - 27.25 \times 27.25}{17.0187}$$

$$= \frac{17.0187}{17.0187} = 1$$

5. **Multiply.** To keep rounding out, we multiply by the swapped matrix first and divide by 17.0187 last:

$$189.8953 \times 12.16 - 27.25 \times 85.2454$$

$$= 2309.127 - 2322.937 = -13.810$$

$$-27.25 \times 12.16 + 4 \times 85.2454$$

$$= -331.360 + 340.982 = 9.622$$

$$\beta_0 = -13.810/17.0187 = -0.811$$

$$\beta_1 = 9.622/17.0187 = 0.565$$

So $\beta_0 = -0.81$, the **intercept** (G-960), and $\beta_1 = 0.57$, the **slope** (G-1823): exactly what the simple linear regression formulas give for these four points. The simple formulas are the **normal equation** (G-1345) with a single feature.

6.2 The picture: a projection

> **Extra:** This section goes beyond the derivation: a second, geometric way to see the same equation.

> **Key point:** Every possible set of predictions lies on one flat plane, and the real targets stick out of it. The best predictions are the point of the plane closest to the targets, where the error meets the plane at a right angle; that right angle gives the **normal equation** (G-1344).

The **normal equation** (G-1345) also has a geometric meaning (ESL §3.2, Figure 3.2; MML §3.8). Treat the n **targets** (G-1949) as one vector y with n entries, and each column of X the same way. A prediction $X\beta$ is a weighted sum of the columns, so all possible predictions fill the flat space the columns span, called the **column space** (G-414) of X. Unless the data lie exactly on a line, y is not in it. The point of the plane straight below y is the **projection** (G-1583) of y onto the plane.

With three **observations** (G-1374) every vector has three entries, so we can draw it. Figure 7 uses the first three students of the worked example: X has the columns $\mathbf{1} = [1, 1, 1]$ and cgpa $= [6.89, 5.12, 7.82]$, and $y = [3.26, 1.98, 3.25]$. Watch the error length as the green point moves through the plane, and the angle at the point where it stops.

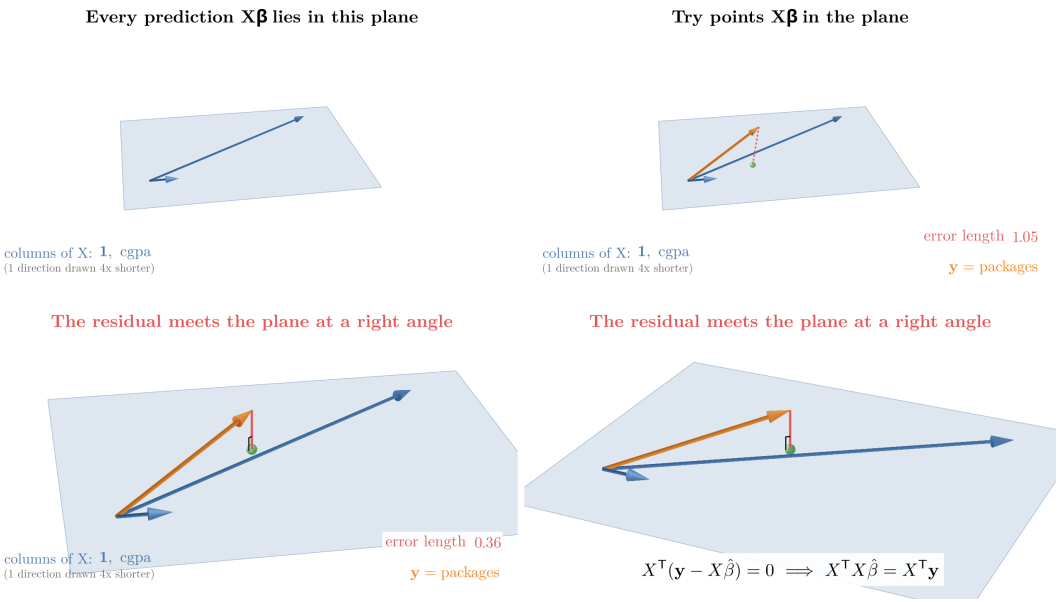


Figure 7: The column space of X for three students is a plane (the two blue columns span it). The target y (orange) sticks out of it. Moving the prediction Xβ through the plane, the error length is smallest, 0.36, at the foot of the perpendicular from y, where the residual meets the plane at a right angle

- **Closest point:** the error $\|y - X\beta\|$ is the length of the dashed line, and its square is E of Section 3. It is smallest, 0.36 (its square, 0.13, is E for these three students), at $\hat{y} = X\hat{\beta} = [2.97, 2.08, 3.44]$, with $\hat{\beta} = [-0.50, 0.50]$.
- **Right angle:** there the **residual** (G-705) [0.29, −0.10, −0.19] is perpendicular to both columns: its **dot product** (G-634) with each column is 0.

The two dot products, one per line:

With the column of 1s:

$$0.29 - 0.10 - 0.19 = 0.00$$

With the cgpa column, one product per line:

$$0.29 \times 6.89 = 1.998$$

$$(-0.10) \times 5.12 = -0.512$$

$$(-0.19) \times 7.82 = -1.486$$

$$1.998 - 0.512 - 1.486 = 0.00$$

Stacked as one equation (the **transpose** (G-2012) of X holds the two columns as rows):

$$X^{\mathsf{T}}(y - X\hat{\beta}) = 0$$

Multiplying out the bracket and moving one term across gives the normal equations of Section 5:

$$X^{\mathsf{T}}X\hat{\beta} = X^{\mathsf{T}}y$$

So the calculus of Section 5 and the right angle of Figure 7 give the same equations. The figure draws the **1** direction four times shorter so that the small residual is visible; shrinking a direction inside the plane does not change the right angle or which point is closest.

7. The cost of the inverse

> **Key point:** Inverting $X^{\mathsf{T}}X$ takes time that grows roughly with the cube of the number of **features** (G-772). With very many features, gradient descent is used instead.

The normal equation has one expensive step: the inverse. The more features, the bigger the **matrix** (G-1180) to invert. $X^{\mathsf{T}}X$ is a square matrix with one row and one column per **coefficient** (G-407), $(m+1) \times (m+1)$. Inverting an $m \times m$ matrix takes on the order of m^3 operations: doubling the features makes the work about 8 times larger.

Figure 8 measures it on this computer.

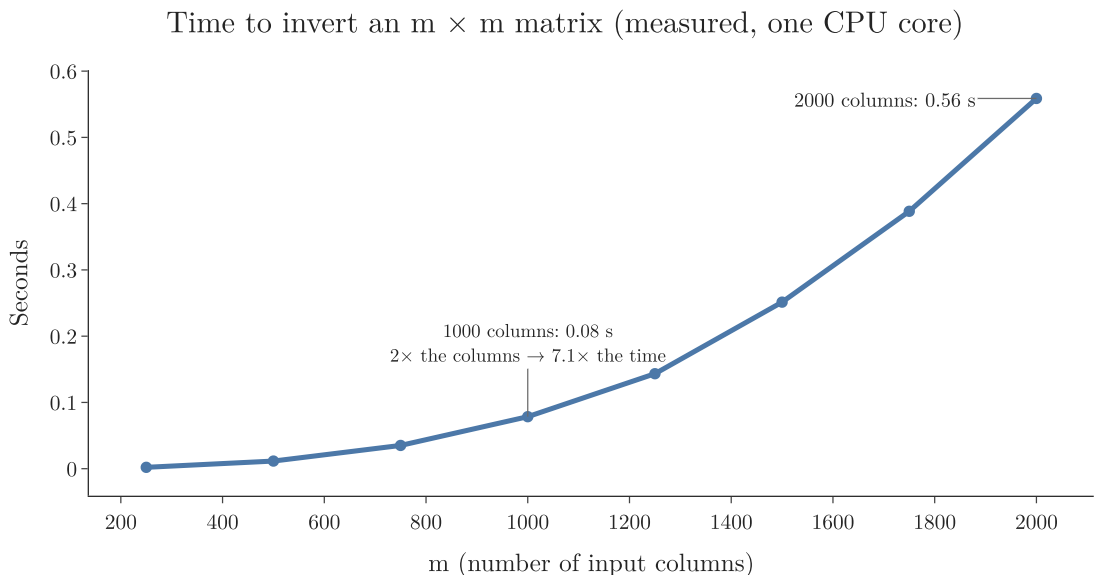


Figure 8: Time to invert a matrix as its size grows

From 1,000 to 2,000 features, the time grows about 7 times, from 0.08 to 0.56 seconds (Figure 8). With tens of thousands of features, as with text or image data, the **inverse** (G-968) becomes very slow and memory-hungry. With numbers, for 20,000 features:

$$20{,}000/1{,}000 = 20 \text{ times the features}$$

$$20^3 = 8{,}000 \text{ times the work}$$

$$8{,}000 \times 0.08 \text{ s} = 640 \text{ s} \approx 11 \text{ minutes}$$

The matrix alone would hold one number per entry, 8 bytes each:

$$20{,}000 \times 20{,}000 \times 8 \text{ bytes} = 3.2 \text{ GB}$$

The cost of the inverse is why there is a second method, **gradient descent** (G-862): it does not compute any inverse, but approaches the best coefficients step by step. Its answer is very close to the **normal equation** (G-1344)'s. In scikit-learn:

- `LinearRegression` uses the closed-form (OLS) solution;
- **`SGDRegressor`** (G-1783) uses gradient descent.

For most tabular data the number of features is small, and `LinearRegression` is the usual choice. **Gradient descent** (G-862) is taught next.

> **Extra:** $X^\top X$ has no inverse when one feature can be built exactly from others (**multicollinearity**, G-1273, as in the dummy variable trap). Then the **normal equation** (G-1345) has no unique answer. Libraries handle this with a "pseudo-inverse" or with regularisation (penalising large coefficients). scikit-learn's `LinearRegression` takes the first route (LAPACK, DGELSD).

8. Summary

Step	Formula	Why it matters
Predictions	$\hat{y} = X\beta$ (X has a first column of 1s)	one product gives every prediction at once
Errors	$e = y - X\beta$	one residual per observation, all in one vector
Sum of squared errors	$E = e^\mathsf{T} e$	a row times a column adds up the squares in one step
Expanded	$E = y^\mathsf{T} y - 2y^\mathsf{T} X\beta + \beta^\mathsf{T} X^\mathsf{T} X\beta$	each of the three parts has a simple derivative rule
Derivative set to zero	$X^\mathsf{T} X\beta = X^\mathsf{T} y$	the error bowl is flat in every direction only at its bottom
Normal equation	$\beta = (X^\mathsf{T} X)^{-1} X^\mathsf{T} y$	multiplying by the inverse is the matrix version of dividing

- A column of 1s in X lets the intercept be treated like any other **coefficient** (G-407), so one formula covers the intercept too.
- One formula gives all $m + 1$ coefficients at once; with one feature it reduces to the simple formulas, so nothing from **simple linear regression** (G-1808) is lost (four students gave the same -0.81 and 0.57 both ways).
- The **inverse** (G-968) costs about m^3 operations, so very wide data uses gradient descent instead (20,000 **features** (G-772) would take about 11 minutes and 3.2 GB for the inverse alone).
- Together these answer the opening question: with matrices, setting the slope of the total squared error to zero in every direction gives one formula for every coefficient, the **normal equation** (G-1344).

9. Sources

Built from

- CampusX, "Multiple Linear Regression | Part 2 | Mathematical Formulation From Scratch", YouTube, https://www.youtube.com/watch?v=NU37mF5q8VE

Other references

- Deisenroth, M. P., Faisal, A. A. and Ong, C. S. (2020). *Mathematics for Machine Learning* (MML). Cambridge University Press. §2.2.2 Inverse and Transpose (the 2 × 2 inverse), §3.8 Orthogonal Projections and §5.5 Useful Identities for Computing Gradients.
- Hastie, T., Tibshirani, R. and Friedman, J. (2009). *The Elements of Statistical Learning*, 2nd ed. (ESL). Springer. §3.2 and Figure 3.2 (the geometry of least squares: y projected onto the column space of X).
- scikit-learn documentation, `sklearn.linear_model.LinearRegression`, Notes section (uses `scipy.linalg.lstsq`).

- LAPACK documentation, DGELSD: minimum-norm least-squares solution using the SVD, for a matrix that may be rank-deficient. netlib.org/lapack.

10. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
Normal equation (G-1344)	A formula that gives all linear-regression coefficients in one step, with no iterations (a closed-form solution): $\beta = (X^\mathsf{T}X)^{-1}X^\mathsf{T}y$.
Inverse matrix (G-968)	The matrix that undoes another: their product is the identity matrix.
Design matrix (X) (G-597)	The data as a matrix, one row per observation and one column per feature, plus a first column of 1s; the 1s multiply the intercept, so it is treated like any other coefficient.
Coefficient vector (β) (G-410)	All the coefficients of the model, β_0 to β_m, as one column.
Transpose (G-2012)	A matrix or vector with rows and columns swapped; it turns a column vector u into a row u^T, so $u^\mathsf{T}x$ is the dot product.
Matrix calculus (G-1176)	Rules for differentiating expressions that contain vectors and matrices, such as a loss written in matrix form, so all the partial derivatives come out in one formula.
Normal equations (G-1345)	A set of equations, one per coefficient, whose solution is the best-fit (least-squares) coefficients: $X^\mathsf{T}X\beta = X^\mathsf{T}y$.
Feature (G-772)	One piece of information about each example that a model uses (e.g. a student's CGPA).
Target (G-1949)	The output column we predict, such as the class; also called the label.
Observation (G-1374)	One record of a dataset: one row of the data table.
Simple linear regression (G-1808)	Linear regression with one input column: it fits a straight line $y = mx + b$ to predict the output from that one feature.
Slope (G-1823)	How much the output changes for one unit of change in the input; m in $y = mx + b$.
Multiple linear regression (G-1279)	Linear regression with several input columns.

Term	Meaning
Intercept (G-960)	The line's value when the input is 0; b in $y = mx + b$.
Coefficient (β_i) (G-407)	The weight of one input column: the change in the output per unit of that input, others fixed.
Matrix (G-1180)	A table of numbers in rows and columns (a 2D tensor); it can hold a dataset (one row per observation), and multiplying a vector by it transforms the vector.
Vector (G-2081)	A list of numbers: a 1D tensor.
Error (residual) (G-705)	The gap between an actual value and the model's prediction on one data point: actual minus predicted.
Residual sum of squares (sum of squared errors, SSE) (G-1684)	The squares of all the errors (residuals) added up; the quantity the best-fit line makes smallest, and a regression tree splits where the SSE of the two sides is smallest.
Partial derivative (G-1457)	The slope of a function of several variables in one variable, holding the others fixed.
Gradient (of the loss) (G-863)	How fast the loss changes with each weight and bias, collected in one vector (the partial derivatives of the loss); it points in the direction in which the loss rises fastest.
Symmetric matrix (G-1932)	A square matrix equal to its own transpose, such as $X^\top X$ or a covariance matrix; it always has a full set of perpendicular eigenvectors.
Identity matrix (G-915)	The matrix that leaves every vector unchanged.
Determinant (G-598)	The factor by which a matrix scales areas (volumes in 3D); 0 when it squishes space into a lower dimension.
Closed-form solution (G-398)	An answer given directly by a formula of ordinary operations.
Ordinary least squares (OLS, least squares) (G-1406)	The way linear regression finds its line with one formula (a closed-form method): the line with the smallest sum of squared errors.
Column space (G-414)	Every output a matrix can produce: all the vectors made by scaling and adding its columns (the span of its columns).
Projection (G-1583)	Dropping a point or vector straight onto an axis, a line or another vector's direction, like casting a shadow.

Term	Meaning
Dot product (G-634)	Multiplying two vectors entry by entry and adding the products, $u^\mathsf{T}x$, which gives one number. It is large when the vectors point the same way, so it measures similarity and gives projections.
Gradient descent (G-862)	Finding the lowest point of a function by repeated small steps downhill.
SGDRegressor (G-1783)	A scikit-learn model that does linear regression with stochastic gradient descent, step by step, so it can also learn from data arriving in small pieces.
Multicollinearity (G-1273)	A relationship between input columns, so that one can be calculated (or largely predicted) from the others; it makes a linear model's coefficients unstable and hard to interpret.